BASE CHANGE FOR A SINGLE COMPLEX UNDER THE EQUIVALENT CONDITIONS, BY INDUCTION AND MAYER-VIETORIS

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Allowed standard prerequisites (editorial).

- Properties of morphisms and isomorphisms of maps between derived pushforwards can be checked locally on the target and extended over unions of quasi-compact opens by the induction principle. invoked at proof lines 6781-6787. Reduces the assertion to affine opens of X and allows the induction over open coverings.
- In the affine case the assertion translates into the module-theoretic condition on the complex, which holds by assumption through the equivalent conditions. invoked at proof lines 6789-6795. Supplies the base case of the induction.
- The relative Mayer-Vietoris distinguished triangles relate the pushforward over a union to the pushforwards over the pieces and their intersection, compatibly with base change. invoked at proof lines 6803-6815. Provides the commutative diagram whose rows are the maps being compared.
- In a commutative diagram of distinguished triangles, if two corresponding horizontal maps are isomorphisms then so is the third. invoked at proof lines 6817-6821. Concludes the induction step from the isomorphisms over U, V and the intersection.

Source reference. Original lemma 0DJ8: sources/stacks-perfect/pinned-original-source.tex, source lines 6654–6654. Cited at proof lines 6790-6794 for the affine base case; the induction and Mayer-Vietoris assembly are inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 6782-6786 to set up the induction; the base case and induction step are inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 6792-6794 for the algebraic translation in the base case; the use of the translated condition is inside the counted span.

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Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Cited at proof lines 6792-6795 for the algebraic translation in the base case; the assembly is inside the counted span.

Source reference. Original lemma 08HZ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 6803-6815 to build the commutative diagram; the diagram and its conclusion are inside the counted span.

Source reference. Original lemma 014A: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 6817-6821 to conclude the induction step; the verification that the other arrows are isomorphisms is inside the counted span.

Source reference. Original lemma 08IB: sources/stacks-perfect/pinned-original-source.tex, source lines 5362–5362. Referenced at proof lines 6771-6774 as the alternative proof method; the actual induction and Mayer-Vietoris proof is inside the counted span.

Lemma 0.2. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram of schemes. Let $K \in D_{QCoh}(\mathcal{O}_X)$ and let $L(g')^*K \to K'$ be a map in $D_{QCoh}(\mathcal{O}_{X'})$. If

(1) *the equivalent conditions of Lemma 0DJ8 hold, and*
(2) *f is quasi-compact and quasi-separated,*

then the composition $Lg^*Rf_*K \to Rf'_*L(g')^*K \to Rf'_*K'$ is an isomorphism.

Proof. We could prove this using the same method as in the proof of Lemma 08IB but instead we will prove it using the induction principle and relative Mayer-Vietoris.

To check the map is an isomorphism we may work locally on S'. Hence we may assume $g : S' \to S$ is a morphism of affine schemes. In particular X is a quasi-compact and quasi-separated scheme. We will use the induction principle of Cohomology of Schemes, Lemma 08DR to prove that for any quasi-compact open $U \subset X$ the similarly constructed map $Lg^*R(U \to S)_*K|_U \to R(U' \to S')_*K'|_{U'}$ is an isomorphism. Here $U' = (g')^{-1}(U)$.

If $U \subset X$ is an affine open, then we find that the result is true by assumption, see Lemma 0DJ8 part (2) and the translation into algebra afforded to us by Lemmas 06Z0 and 08DW.

The induction step. Suppose that $X = U \cup V$ is an open covering with U, V, $U \cap V$ quasi-compact such that the result holds for U, V, and $U \cap V$. Denote $a = f|_U$, $b = f|_V$ and $c = f|_{U \cap V}$. Let $a' : U' \to S'$, $b' : V' \to S'$ and $c' : U' \cap V' \to S'$ be the base changes of a, b, and c. Using the distinguished triangles from relative

Mayer-Vietoris (Cohomology, Lemma 08HZ) we obtain a commutative diagram

$$\begin{array}{ccc}
Lg^*Rf_*K & \longrightarrow & Rf'_*K' \\
\downarrow & & \downarrow \\
Lg^*Ra_*K|_U \oplus Lg^*Rb_*K|_V & \longrightarrow & Ra'_*K'|_{U'} \oplus Rb'_*K'|_{V'} \\
\downarrow & & \downarrow \\
Lg^*Rc_*K|_{U\cap V} & \longrightarrow & Rc'_*K'|_{U'\cap V'} \\
\downarrow & & \downarrow \\
Lg^*Rf_*K[1] & \longrightarrow & Rf'_*K'[1]
\end{array}$$

Since the 2nd and 3rd horizontal arrows are isomorphisms so is the first (Derived Categories, Lemma 014A) and the proof of the lemma is finished. □